# Notes: Kacperczyk, Van Nieuwerburgh, and Veldkamp (JF, 2014)

### Time-Varying Fund Manager Skill

## Contents

# 1 Big picture

- **Question.** Do active mutual fund managers add value, and if so, is their skill a fixed talent such as "stock picking" or "market timing," or is it a more general ability whose application changes with macroeconomic conditions?
- **Setting.** The paper studies the universe of actively managed U.S. domestic equity mutual funds from 1980 to 2005, combining CRSP mutual fund data, Thomson Reuters holdings, CRSP stock returns, and a matched manager-level panel.
- **Core challenge.** Standard performance decompositions usually look for stock picking and market timing unconditionally. If market timing shows up mainly in recessions, and recessions are rare, then unconditional tests can easily miss it.
- **Main idea.** Skill is modeled conceptually as general cognitive ability to process information. In booms, skilled managers optimally spend more effort on firm-specific information and therefore look like stock pickers. In recessions, they spend more effort on aggregate information and therefore look like market timers.
- **Main empirical strategy.** Use holdings-based measures of *Timing* and *Picking*. Timing captures whether a fund overweights high-beta stocks before strong market returns; Picking captures whether a fund overweights stocks before strong idiosyncratic returns. Regress these measures on recession indicators, study the upper tail of the skill distribution, test whether the *same* funds switch strategies over the business cycle, and then build a real-time skill index.
- **Main result.** Mutual funds become better market timers in recessions and relatively better stock pickers in expansions. In the baseline regression, the recession coefficient is about $+0.14$ percentage points per month for Timing and about $-0.15$ percentage points per month for Picking, meaning that the use of skill rotates strongly toward timing in bad times.
- **Key cross-sectional result.** The cyclical shift is strongest among the most skilled funds. Funds that are top stock pickers in expansions are also top market timers in recessions. This is the central "same manager, two tasks" result.
- **Performance result.** These strategy-switching funds outperform other funds by about 50 to 80 basis points per year before expenses and on a risk-adjusted basis.
- **Skill-index result.** A real-time *Skill Index* that weights market timing more in recessions and stock picking more in booms is much more persistent than either Timing or Picking alone. A one-standard-deviation increase in the index predicts about 2.3% higher return performance over the next year net of expenses and about 1.1% higher performance after richer factor controls.
- **Why this matters.** The paper changes the interpretation of the mutual-fund literature. Weak unconditional timing evidence does not mean market timing skill is absent. It may mean that timing is state-dependent.
- **Economic takeaway.** The paper is not saying that managers have one immutable type of skill. It argues instead that skilled managers reallocate attention across tasks over the business cycle, and that this time-varying use of skill creates value.

# 2 Data and measurement (key objects)

## 2.1 Data sources and sample construction

- **Fund-level data.** The starting point is the CRSP Survivorship-Bias-Free Mutual Fund Database, which provides fund returns and characteristics such as assets under management, age, expenses, turnover, and load.
- **Holdings data.** The fund data are merged with the Thomson Reuters holdings database and with CRSP stock returns using the Kacperczyk–Sialm–Zheng matching procedure.
- **Manager data.** The authors match funds to manager names using CRSP, Morningstar, Nelson's Directory of Investment Managers, Zoominfo, and Zabasearch.
- **Stock-level data.** CRSP/Compustat provide stock returns, market capitalizations, book-to-market ratios, and momentum variables.

## 2.2 Sample restrictions

- The focus is on domestic open-end diversified equity mutual funds.
- The paper excludes index funds, sector funds, and funds with less than 80% of the portfolio in stocks.
- Multiple share classes are aggregated to the fund level because they share the same portfolio.
- To reduce incubation bias, the authors drop observations before the reported fund start year, funds with missing names, funds with less than $5 million in lagged assets under management, and funds with fewer than 10 stocks.

## 2.3 Sample size and time span

- The resulting sample contains 3,477 distinct funds and 250,219 fund-month observations.
- The matched manager sample contains 4,267 managers.
- The sample runs monthly from January 1980 to December 2005, for a total of 312 months.
- Of these 312 months, 38 are NBER recession months, or about 12% of the sample.

## 2.4 What is observed and what is not

- **Observed directly.** Fund returns, portfolio holdings, stock-level returns, fund characteristics, and manager characteristics.
- **Not observed directly.** Manager skill itself, effort allocation across firm-specific versus aggregate information, and the true information set used by the manager when the portfolio was formed.
- **Implication.** The paper has to infer skill from how portfolio weights line up with later realizations of systematic and idiosyncratic returns.

### 2.5 Recession indicators and controls

- **Baseline recession measure.** The NBER recession indicator.
- **Real-time alternatives.**
  - `RecRT`: the Chauvet–Piger real-time recession probability.
  - `RecCFNAI`: minus the Chicago Fed National Activity Index.
- **Fund controls.** The main regressions include log age, log total net assets, expense ratio, turnover, flow, load, and portfolio style controls for size, value, and momentum.
- **Important timing convention.** These controls are demeaned, so the intercept in the main regressions is interpreted as the level of skill in expansions, while the recession coefficient measures the change in skill in recessions.

### 2.6 Why holdings matter

- Holdings allow the authors to look through realized fund returns and ask *what kind of risk the manager chose to bear* before returns were realized.
- This is crucial because unconditional alpha mixes together stock picking, market timing, fee policy, and passive factor exposures.
- The paper's main objects, Timing and Picking, are best thought of as holdings-based hypothetical returns.

## 3 Models and empirical specifications (all equations + interpretation)

### 3.1 Timing and Picking as holdings-based skill measures

For fund $j$ in month $t$, the paper defines market-timing skill as

$$\text{Timing}_t^j = \sum_{i=1}^{N^j} \left(w_{i,t}^j - w_{i,t}^m\right) \left(\beta_{i,t} R_{t+1}^m\right), \tag{1}$$

and stock-picking skill as

$$\text{Picking}_t^j = \sum_{i=1}^{N^j} \left(w_{i,t}^j - w_{i,t}^m\right) \left(R_{i,t+1} - \beta_{i,t} R_{t+1}^m\right). \tag{2}$$

**Objects.**

- $w_{i,t}^j$ is fund $j$'s beginning-of-period portfolio weight in stock $i$.
- $w_{i,t}^m$ is the stock's market-capitalization weight in the market portfolio.
- $\beta_{i,t}$ is estimated from a 12-month rolling-window regression of the stock's excess return on the market's excess return.

- $R^m_{t+1}$ is the realized market return between $t$ and $t+1$.

**Interpretation.**

- Timing asks: before the market moved, did the manager overweight high-beta stocks when market returns were about to be high and underweight them when market returns were about to be low?
- Picking asks: before stock-specific shocks were realized, did the manager overweight the stocks that were about to earn high idiosyncratic returns?
- Both measures are expressed in units of return per month and are *hypothetical* returns based on beginning-of-period holdings.
- The measures are unconditional in the sense that they do not strip out the public-information conditioning set as in Ferson–Schadt type timing measures. The paper wants the skill concept to include processing public *or* private information.

## 3.2 Main cyclical-skill regressions

The baseline tests are panel regressions of the form

$$\text{Picking}_t^j = a_0 + a_1 \,\text{Recession}_t + a_2 X_t^j + \varepsilon_t^j, \tag{3}$$

$$\text{Timing}_t^j = b_0 + b_1 \,\text{Recession}_t + b_2 X_t^j + \eta_t^j. \tag{4}$$

**Interpretation.**

- $a_1 < 0$ means the use of stock-picking skill falls in recessions.
- $b_1 > 0$ means the use of market-timing skill rises in recessions.
- Because all controls are demeaned, $a_0$ and $b_0$ can be read as expansion values, and $a_1$ and $b_1$ as recession-minus-expansion shifts.
- Standard errors are clustered by fund and time because recession is a common monthly shock across all funds.

**What Table I says.**

- With controls, the recession coefficient is about $+0.139$ for Timing and $-0.146$ for Picking.
- That means that relative to expansions, timing is about 14 basis points per month higher in recessions, while picking is about 15 basis points per month lower.
- The constants are near zero for Timing and slightly negative for Picking, which is consistent with the fact that the *average* active fund does not beat passive benchmarks.

## 3.3 Real-time recession indicators

The same specifications are re-estimated with two real-time macro indicators.

**Interpretation.**

- This matters because NBER recessions are dated ex post. Managers do not observe the NBER chronology in real time.
- Using `RecRT`, moving from a 0% to a 50% recession probability raises Timing by about 20 basis points per month and lowers Picking by about 12 basis points per month.
- Using `RecCFNAI`, moving to a recessionary reading of 0.7 raises Timing by about 7 basis points per month and lowers Picking by about 4 basis points per month.
- So the basic pattern is not an artifact of ex post business-cycle dating.

### 3.4 Quantile regressions: who changes behavior most?

The paper next estimates the same models at different quantiles of the cross-sectional distribution of skill.

**Interpretation.**

- If all managers merely drift a little toward timing in recessions, then the recession coefficient should look similar at the median and in the right tail.
- If only the best managers really reallocate attention across tasks, then the effect should be much stronger in the right tail.

**What Table III says.**

- At the median, the recession effect on Timing is about 0.059.
- At the 95th percentile, it is about 0.251.
- At the median, the recession effect on Picking is about −0.084.
- At the 95th percentile, it is about −0.173.

So the managers in the upper tail of the skill distribution change the use of their skills much more strongly over the business cycle than the typical manager.

### 3.5 Testing whether the same funds do both tasks

The paper constructs a `Top` indicator for the funds with the strongest stock-picking record in expansions. More precisely, it identifies the 25% of funds with the largest fraction of expansion months spent in the top quartile of the cross-sectional Picking distribution. It then estimates, separately by state,

$$\text{Ability}_t^j = c_0 + c_1\,\text{Top}_j + c_2 X_t^j + \nu_t^j, \tag{5}$$

where Ability is either Timing or Picking.

**Interpretation.**

- If the same funds truly switch tasks, then the funds that are superior pickers in expansions should also be superior timers in recessions.
- If instead some funds are “pickers” and other funds are “timers,” then there is no reason for this to hold.

**What Table IV says.**

- In expansions, the `Top` group has Picking about 0.059 higher than other funds, which is true by construction.
- In recessions, the same `Top` group has Timing about 0.037 higher than other funds.
- The `Top` group does *not* have better Timing in expansions and does *not* have better Picking in recessions.

This is the cleanest piece of evidence that skilled managers are not just permanently endowed with one task-specific talent. They switch.

## 3.6 Fund skill or manager skill?

The paper reruns the cyclical regressions at the manager level and then adds manager fixed effects.

**Interpretation.**

- If the cyclical effect merely reflects organizational capital or fund-family structure, it should weaken once one tracks the same manager over time.
- If the effect survives with manager fixed effects, it is much more natural to interpret it as manager skill.

**What Table V says.**

- At the manager level, the recession coefficient is about $+0.156$ for Timing and $-0.192$ for Picking.
- With manager fixed effects, the coefficients remain about $+0.160$ and $-0.187$.

So the cyclical shift is not driven by turnover in the population of managers. The same manager displays different skill manifestations across states of the business cycle.

## 3.7 How do managers time the market?

The paper then opens the black box of Timing.

**Cash channel.**

- In expansions, funds hold about 5.25% of the portfolio in cash.
- In recessions, implied cash rises by about 3 percentage points and reported cash rises by about 0.4 percentage points.
- Month-over-month, implied cash rises by about 0.5 percentage points in recessions but falls by about 1.5 percentage points in expansions.

**Beta channel.**

- The value-weighted equity beta of fund holdings is about 1.11 in expansions and 1.00 in recessions.
- This means managers hold lower-beta stocks in recessions.

**Sector-rotation channel.**

- In recessions, funds tilt toward defensive, lower-beta sectors such as Healthcare, Nondurables, Wholesale, and Utilities.
- They tilt away from higher-beta sectors such as Telecom, Business Equipment and Services, Manufacturing, Energy, and Durables.

**Interpretation.** Timing is not a mysterious residual. Managers time the market by holding more cash, lowering portfolio beta, and rotating across industries in recessionary periods.

### 3.8 Alternative explanations and the herding test

The paper considers a series of alternative explanations.

**Composition effects.**

- Fund fixed effects leave the main recession coefficients essentially unchanged.
- The same logic holds at the manager level with manager fixed effects.
- Observable manager characteristics do not shift systematically over the business cycle in a way that could explain the results.

**Mechanical stock-return effects.**

- The authors simulate panels of stocks and artificial funds with time-varying CAPM moments.
- Random or mechanically tilted portfolio strategies do not reproduce the empirical pattern of higher Timing in recessions and higher Picking in expansions.

**Career-concerns / herding explanation.**

- If managers herd more in recessions, they may look more like market timers without being more skilled.
- To test this, the paper defines portfolio dispersion as

$$\text{PortfolioDispersion}_t^j = \sum_{i=1}^{N^j} \left( w_{i,t}^j - w_{i,t}^m \right)^2 . \tag{6}$$

- Higher dispersion means *less* herding.
- The recession coefficient in this regression is about 0.347, significant at the 5% level.

So managers herd *less*, not more, in recessions. This goes against the career-concerns story.

### 3.9 Real-time Skill Index

The paper's second major contribution is a real-time measure of general managerial ability:

$$\text{SkillIndex}_{t+1}^j = w_t \, \text{Timing}_t^j + (1 - w_t) \, \text{Picking}_t^j , \tag{7}$$

where $w_t$ is the real-time recession probability from Chauvet and Piger (2008).

**Construction details.**

- Timing and Picking are normalized each month to have cross-sectional mean zero and standard deviation one.
- In recessions, $w_t$ is high, so the index puts more weight on Timing.
- In expansions, $w_t$ is low, so the index puts more weight on Picking.
- The index can be computed at time $t+1$ using only information known by then, so it is investor-usable in real time.

**Interpretation.** The index is designed to capture general cognitive ability rather than a fixed single skill. A skilled manager should look good on the dimension that matters most in the current macro state.

### 3.10 Future alpha construction and predictive regressions

The paper evaluates whether the Skill Index predicts future performance using rolling-window factor models. For example, the one-month-ahead CAPM alpha is based on

$$R^j_{t+2} = \alpha^j + \beta^j R^m_{t+2} + \varepsilon^j_{t+2}, \tag{1}$$

with the 12-month estimation window running from $t-10$ to $t+2$. The realized abnormal component is then

$$\widehat{\alpha}^j_{t+2} = R^j_{t+2} - \widehat{\beta}^j R^m_{t+2}, \tag{2}$$

and analogous constructions are used for three-factor and four-factor alphas.

The predictive regressions then ask whether

$$\text{Future Alpha}^j_{t+h} = d_0 + d_1\,\text{SkillIndex}^j_{t+1} + d_2 X^j_t + u^j_{t+h}, \tag{3}$$

for one-month-ahead and one-year-ahead horizons.

**Interpretation.**

- This is the investor-facing exercise: can a real-time measure of state-dependent skill help identify better funds going forward?
- The paper does both portfolio sorts and predictive panel regressions because each has different strengths. Sorts avoid overlapping-data concerns; predictive regressions allow rich controls.

## 4 Identification and instruments (what makes the estimates credible)

### 4.1 What identifies the paper's main claim

- There is no instrumental-variables design here. Identification comes from a structured holdings-based decomposition combined with state variation and a battery of within-manager and within-fund tests.
- The first source of identification is the distinction between systematic and idiosyncratic components of stock returns in equations (1) and (2).

- The second is time variation in the business cycle. If managers truly reallocate attention across tasks, the same holdings-based measure should tilt toward timing in recessions and toward picking in booms.
- The third is cross-sectional heterogeneity: the effect should be stronger in the upper tail of skill, not just at the median.

### 4.2 Why unconditional fund-performance tests can miss the result

- Recessions are only 38 of 312 months in the sample.
- If timing skill shows up primarily in recessions, averaging all months together makes timing look weak or absent.
- This is why the paper's core contribution is not merely a new variable; it is a new conditioning logic.

### 4.3 Why the same-fund and same-manager tests matter

- The `Top` analysis in Table IV rules out a simple "some funds are timers, other funds are pickers" interpretation.
- The manager fixed-effect analysis in Table V rules out a pure composition story in which recessions simply change which people happen to be managing money.
- Together, these two exercises make the paper's interpretation much more compelling: it is the same manager reallocating effort across tasks.

### 4.4 Why the real-time recession indicators matter

- The baseline NBER measure is useful for ex post evaluation, but managers do not observe the NBER chronology in real time.
- The fact that the same pattern appears with the Chauvet–Piger recession probability and with CFNAI means the result is not just a retrospective labeling exercise.
- This also justifies the construction of the real-time Skill Index.

### 4.5 What makes the results credible

- The paper attacks the most obvious non-skill explanations directly:
  - fund composition,
  - manager composition,
  - size-driven composition,
  - purely mechanical stock-return effects,
  - career-concerns-induced herding.
- Each explanation predicts a different diagnostic pattern, and those patterns do not line up with the data.

- The manager-level and out-of-sample real-time results are especially important because they connect the decomposition exercises to economically meaningful future performance.

### 4.6 Main limitations

- The paper is highly persuasive, but it is still not a clean quasi-experiment. The business cycle is not randomly assigned.
- Holdings are observed at disclosure frequency, not literally continuously, so very high-frequency tactical changes are not visible.
- Timing and Picking are noisy realized-skill proxies. A manager can be skilled but unlucky over short windows.
- The skill index depends on the choice of real-time recession proxy and on the linear weighting scheme in equation (7), even though the paper shows robustness to alternatives.
- The paper documents value creation before expenses and after factor adjustment, but the investability of the signal in real time can still be attenuated by search frictions, trading costs, and learning by investors.

### 4.7 Relation to the literature

- Relative to the classic mutual-fund literature, the paper says the right decomposition is not unconditional picking versus timing, but state-contingent picking versus timing.
- Relative to Ferson–Schadt type conditional-performance papers, the paper uses unconditional holdings-based measures and interprets skill more broadly as information processing, not merely beating a public-information benchmark.
- Relative to the broader business-cycle fund literature, the paper's distinctive result is that the *same* funds do different valuable things in different states.
- The paper also ties naturally to the authors' related rational-attention theory, which provides an explicit mechanism for why aggregate information becomes relatively more valuable in recessions.

## 5 Tables and figures

**General comment.** The paper is table-driven. The early tables establish the cyclical shift in the use of skill, the middle tables show that the same funds and managers are responsible for it, and the last tables show that the real-time skill index predicts future performance. The one main figure is a persistence figure.

### 5.1 Table I and Table II: the basic cyclical facts

**Read:**

- Table I is the paper's baseline fact. In the controlled specification, recessions raise Timing by about 0.139 and lower Picking by about 0.146.

**Table 1**
**Summary statistics**

| | Mean | Median | Standard deviation |
|---|---|---|---|
| Number of distinct mutual funds | 2543 | | |
| Number of fund-month observations | 211,001 | | |
| Number of funds per month | 879 | 720 | |
| Proportion of index funds (in %) | 4.53 | | |
| Proportion of load funds (in %) | 54.22 | | |
| TNA (total net assets) (in millions) | 952 | 166 | 3,771 |
| Age | 13.49 | 8 | 13.98 |
| Expense ratio (in %) | 1.24 | 1.20 | 0.44 |
| Turnover ratio (in %) | 88.06 | 65.00 | 103.51 |
| Mean of prior-year new money growth (in % per month; winsorized) | 2.50 | 0.35 | 9.45 |
| Mean investor return (in % per month) | 0.85 | 1.15 | 5.79 |
| Standard deviation of investor returns over prior year (in % per month) | 5.27 | 4.87 | 2.48 |
| Proportion invested in stocks (in %) | 93.16 | 95.22 | 7.72 |
| Proportion invested in cash (in %) | 5.51 | 3.81 | 6.51 |
| Proportion Invested in bonds (in %) | 0.75 | 0 | 2.55 |
| Proportion invested in preferred stocks (in %) | 0.24 | 0 | 1.91 |
| Proportion invested in other securities (in %) | 0.33 | 0 | 2.60 |
| Difference in TNA after adjusting for nonstock holdings (in %) | 8.33 | 3.73 | 17.64 |
| Trading costs per year (in %) | 0.58 | 0.36 | 0.66 |
| Weight of recent IPOs divided by length of disclosure period (in %) | 0.22 | 0.01 | 0.49 |
| Correlation between holdings and investor returns (in %) | 97.96 | 99.11 | 5.06 |
| Value of trades relative to market capitalization (in %) | 0.28 | 0.11 | 0.45 |
| Size score (score ranging between 1–5 using size quintiles) | 4.05 | 4.44 | 0.97 |
| Value score (score ranging between 1–5 using book-to-market quintiles) | 2.58 | 2.57 | 0.51 |
| Momentum score (score ranging between 1–5 using momentum quintiles) | 3.33 | 3.29 | 0.61 |

This table presents the summary statistics for the sample of equity mutual funds over the period 1984 to 2003.

**Table 2**
**Performance of investor and holdings returns**

| | Investor return | Holdings return | Return gap |
|---|---|---|---|
| Panel A: Equal-weighted returns | | | |
| Raw return | $1.014^{***}$ | $1.003^{***}$ | 0.011 |
| | (0.305) | (0.305) | (0.009) |
| CAPM alpha | −0.064 | −0.077 | 0.012 |
| | (0.056) | (0.056) | (0.010) |
| Fama–French alpha | −0.057 | −0.062 | 0.005 |
| | (0.044) | (0.045) | (0.009) |
| Carhart alpha | −0.068 | −0.071 | 0.002 |
| | (0.045) | (0.046) | (0.009) |
| Panel B: Value-weighted returns | | | |
| Raw return | $0.988^{***}$ | $0.998^{***}$ | −0.010 |
| | (0.294) | (0.295) | (0.012) |
| CAPM alpha | $-0.075^{**}$ | $-0.067^{**}$ | −0.009 |
| | (0.032) | (0.033) | (0.012) |
| Fama–French alpha | $-0.064^{**}$ | −0.045 | $-0.019^{*}$ |
| | (0.031) | (0.032) | (0.011) |
| Carhart alpha | $-0.072^{**}$ | −0.051 | $-0.021^{*}$ |
| | (0.032) | (0.033) | (0.012) |

This table summarizes the monthly investor returns, the holdings returns after subtracting expenses, and the return gaps for the equal- and value-weighted portfolio of all funds in our sample over the period 1984 to 2003. The return gap has been defined as the difference between the investor return and the holdings return of the portfolio disclosed in the previous period. The holdings return is reported after subtracting fund expenses. We report the raw returns, the one-factor alpha of Jensen (1968), the three-factor alpha of Fama and French (1993), and the four-factor alpha of Carhart (1997). The returns are expressed in percent per month and the standard errors are summarized in parentheses.The significance levels are denoted by *, **, and *** and indicate whether the results are statistically different from zero at the 10-, 5-, and 1-percent significance levels.

- Economically, that means roughly 1.67% per year more timing ability in recessions and roughly 1.75% per year less picking ability than in expansions.
- Table II shows the same pattern with real-time recession measures.
- Moving the Chauvet–Piger recession probability from 0% to 50% raises Timing by about 20 basis points per month and lowers Picking by about 12 basis points per month.
- These are the tables to remember if someone asks, "Does the kind of skill change over the cycle?"

## 5.2 Table III and Table IV: only the best funds really switch tasks

**Read:**

- Table III shows that the recession effect is much larger in the upper tail of the skill distribution than at the median.
- At the 95th percentile, the recession effect on Timing is about four times the median effect.
- Table IV is the key same-fund table. Funds selected for superior stock picking in expansions also have superior market timing in recessions.
- The `Top` funds have about 0.059 more Picking in expansions and about 0.037 more Timing in recessions, but no corresponding advantage in the "wrong" state-task pairings.
- This is where the paper moves from "average funds tilt with the cycle" to "the same skilled funds switch strategies."

**Table 3**
**Persistence of the return gap**

| | Raw return gap | | | Abnormal return gap using four-factor model | | |
|---|---|---|---|---|---|---|
| | 1 Year | 3 Years | 5 Years | 1 Year | 3 Years | 5 Years |
| 1. Decile | −0.113*** | −0.122*** | −0.113*** | −0.088*** | −0.113*** | −0.100*** |
| (lowest RG) | (0.026) | (0.024) | (0.027) | (0.025) | (0.022) | (0.026) |
| 2. Decile | −0.026 | −0.040*** | −0.055*** | −0.015 | −0.041*** | −0.063*** |
| | (0.020) | (0.014) | (0.020) | (0.020) | (0.015) | (0.021) |
| 3. Decile | −0.019 | −0.024 | −0.028** | −0.017 | −0.024 | −0.028** |
| | (0.011) | (0.022) | (0.013) | (0.011) | (0.023) | (0.013) |
| 4. Decile | −0.015 | −0.016 | −0.022* | −0.024*** | −0.018 | −0.018 |
| | (0.010) | (0.012) | (0.012) | (0.009) | (0.012) | (0.012) |
| 5. Decile | −0.008 | −0.002 | −0.010 | −0.015 | −0.012 | −0.013 |
| | (0.009) | (0.011) | (0.012) | (0.010) | (0.012) | (0.012) |
| 6. Decile | 0.001 | −0.001 | −0.006 | −0.013 | −0.009 | −0.013 |
| | (0.011) | (0.010) | (0.012) | (0.011) | (0.010) | (0.012) |
| 7. Decile | −0.003 | −0.013 | −0.005 | −0.016 | −0.024** | −0.019 |
| | (0.012) | (0.012) | (0.014) | (0.012) | (0.012) | (0.014) |
| 8. Decile | 0.018 | 0.029** | 0.020 | 0.002 | 0.013 | −0.004 |
| | (0.014) | (0.012) | (0.017) | (0.014) | (0.012) | (0.016) |
| 9. Decile | 0.049*** | 0.053*** | 0.050* | 0.025* | 0.023 | 0.040 |
| | (0.016) | (0.019) | (0.028) | (0.015) | (0.018) | (0.030) |
| 10. Decile | 0.154*** | 0.111*** | 0.122*** | 0.116*** | 0.078*** | 0.088*** |
| (highest RG) | (0.033) | (0.025) | (0.023) | (0.031) | (0.023) | (0.022) |
| Decile 10— | 0.268*** | 0.234*** | 0.235*** | 0.204*** | 0.191*** | 0.188*** |
| Decile 1 | (0.043) | (0.031) | (0.033) | (0.040) | (0.028) | (0.031) |
| Second half— | 0.080*** | 0.077*** | 0.082*** | 0.055*** | 0.058*** | 0.063*** |
| First half | (0.016) | (0.012) | (0.012) | (0.015) | (0.011) | (0.012) |
| Spearman | 0.988*** | 0.964*** | 1.000*** | 0.839*** | 0.906*** | 0.924*** |
| correlation | (0.000) | (0.000) | (0.000) | (0.002) | (0.000) | (0.000) |

This table reports the means and the standard errors (in parentheses) of the monthly return gaps for decile portfolios of mutual funds sorted by their average lagged return gaps during the previous 1, 3, and 5 years over the period 1984 to 2003. The return gap is defined as the difference between the reported return and the holdings return of the portfolio disclosed in the previous period. The first three columns summarize the raw return gaps, and the last three columns show the four-factor abnormal return gaps following Carhart (1997). The returns are expressed in percent per month. The table also calculates the differences in the return gaps between the top and the bottom deciles and the top and the bottom halves, along with the Spearman rank correlations and the corresponding *p*-values in parentheses. The significance levels are denoted by *, **, and *** and indicate whether the results are statistically different from zero at the 10-, 5-, and 1-percent significance levels.

**Table 4**
**Portfolio returns based on the return gap**

| | Excess return | CAPM alpha | Fama French alpha | Carhart alpha | Pástor Stambaugh alpha | Ferson Schadt alpha | DGTW selectivity measure | GT performance measure |
|---|---|---|---|---|---|---|---|---|
| 1. Decile: | −0.183* | −0.246** | −0.164*** | −0.199*** | −0.196*** | −0.191*** | 0.065 | 0.195* |
| Mean: −0.598 | (0.098) | (0.095) | (0.061) | (0.062) | (0.062) | (0.061) | (0.061) | (0.107) |
| 2. Decile | −0.090 | −0.118* | −0.110** | −0.123** | −0.110** | −0.093* | 0.050 | 0.124 |
| Mean: −0.245 | (0.062) | (0.061) | (0.053) | (0.054) | (0.052) | (0.050) | (0.043) | (0.087) |
| 3. Decile | −0.064 | −0.051 | −0.082* | −0.061 | −0.047 | −0.063* | 0.054 | 0.126* |
| Mean: −0.137 | (0.051) | (0.052) | (0.048) | (0.049) | (0.046) | (0.037) | (0.037) | (0.072) |
| 4. Decile | −0.062 | −0.044 | −0.084* | −0.066 | −0.052 | −0.073** | 0.045 | 0.090 |
| Mean: −0.070 | (0.049) | (0.049) | (0.047) | (0.048) | (0.046) | (0.035) | (0.033) | (0.063) |
| 5. Decile | −0.066 | −0.032 | −0.090* | −0.059 | −0.042 | −0.067* | 0.033 | 0.066 |
| Mean: −0.019 | (0.053) | (0.052) | (0.049) | (0.050) | (0.046) | (0.035) | (0.033) | (0.058) |
| 6. Decile | −0.018 | 0.013 | −0.032 | −0.012 | 0.003 | −0.027 | 0.053* | 0.075 |
| Mean: 0.026 | (0.051) | (0.049) | (0.048) | (0.049) | (0.046) | (0.033) | (0.031) | (0.060) |
| 7. Decile | −0.053 | −0.037 | −0.080 | −0.069 | −0.049 | −0.077 * | 0.042 | 0.127* |
| Mean: 0.078 | (0.058) | (0.058) | (0.056) | (0.057) | (0.052) | (0.041) | (0.039) | (0.068) |
| 8. Decile | −0.064 | −0.063 | −0.086* | −0.083 | −0.065 | −0.087** | 0.025 | 0.110 |
| Mean: 0.149 | (0.058) | (0.059) | (0.051) | (0.052) | (0.048) | (0.040) | (0.039) | (0.077) |
| 9. Decile | 0.029 | −0.003 | 0.022 | −0.019 | −0.008 | 0.026 | 0.091* | 0.200** |
| Mean: 0.266 | (0.082) | (0.082) | (0.056) | (0.056) | (0.055) | (0.052) | (0.048) | (0.098) |
| 10. Decile: | 0.101 | 0.012 | 0.156** | 0.025 | 0.026 | 0.068 | 0.125* | 0.322** |
| Mean: 0.657 | (0.151) | (0.148) | (0.078) | (0.071) | (0.071) | (0.072) | (0.075) | (0.140) |
| Decile 10— | 0.284*** | 0.259*** | 0.321*** | 0.225*** | 0.222*** | 0.258*** | 0.060* | 0.127** |
| Decile 1 | (0.078) | (0.078) | (0.059) | (0.054) | (0.054) | (0.053) | (0.038) | (0.055) |
| Second half— | 0.092*** | 0.083*** | 0.102*** | 0.070*** | 0.071*** | 0.078*** | 0.018 | 0.047** |
| First half | (0.032) | (0.032) | (0.025) | (0.024) | (0.024) | (0.023) | (0.015) | (0.021) |
| Spearman | 0.839*** | 0.697** | 0.794*** | 0.649** | 0.685** | 0.661** | 0.103 | 0.297 |
| correlation | (0.002) | (0.025) | (0.006) | (0.042) | (0.029) | (0.038) | (0.770) | (0.405) |

This table reports the mean monthly returns and the corresponding standard errors (in parentheses) for deciles of mutual funds sorted according to the lagged 1-year return gap over the period 1984 to 2003. The return gap is lagged for one additional quarter to account for the possible delay in reporting the holdings. The return gap is defined as the difference between the investor fund return and the return based on the previous holdings. In the first column we report the mean lagged return gap for the decile portfolios. We use the excess return over the market, the one-factor alpha of Jensen (1968), the three-factor alpha of Fama and French (1993), the four-factor alpha of Carhart (1997), the five-factor model of Pástor and Stambaugh (2003), and the Ferson and Schadt (1996) conditional measure based on the four-factor model to measure fund performance. Moreover, we report the Characteristic Selectivity (CS) measure of Daniel, Grinblatt, Titman, and Wermers (1997), and the Grinblatt and Titman (1993) performance measure. The returns are expressed in percent per month. The table also reports the differences in the return gaps between the top and the bottom deciles and the top and the bottom halves, along with the Spearman rank correlations and the corresponding *p*-values in parentheses. The significance levels are denoted by *, **, and *** and indicate whether the results are statistically different from zero at the 10-, 5-, and 1-percent significance levels.

**Table 5**
**Portfolio returns based on the return gap with back-testing**

| | Excess return | CAPM alpha | Fama French alpha | Carhart alpha | Pástor Stambaugh alpha | Ferson Schadt alpha | DGTW selectivity measure | GT performance measure |
|---|---|---|---|---|---|---|---|---|
| 1. Decile: | −0.366*** | −0.431*** | −0.396*** | −0.328*** | −0.310*** | −0.195** | −0.027 | 0.071 |
| Mean: −0.600 | (0.106) | (0.103) | (0.098) | (0.099) | (0.098) | (0.097) | (0.070) | (0.107) |
| 2. Decile | −0.243*** | −0.278*** | −0.279*** | −0.209** | −0.190** | −0.048 | −0.012 | 0.066 |
| Mean: −0.245 | (0.088) | (0.088) | (0.088) | (0.089) | (0.086) | (0.085) | (0.052) | (0.090) |
| 3. Decile | −0.166** | −0.175** | −0.182** | −0.098 | −0.078 | 0.005 | 0.043 | 0.092 |
| Mean: −0.137 | (0.084) | (0.085) | (0.088) | (0.086) | (0.083) | (0.079) | (0.053) | (0.092) |
| 4. Decile | −0.186** | −0.187** | −0.202*** | −0.127* | −0.110 | −0.001 | 0.016 | 0.036 |
| Mean: −0.070 | (0.074) | (0.075) | (0.077) | (0.076) | (0.074) | (0.069) | (0.044) | (0.081) |
| 5. Decile | −0.047 | −0.040 | −0.054 | 0.011 | 0.033 | 0.104 | 0.056 | 0.022 |
| Mean: −0.019 | (0.069) | (0.070) | (0.072) | (0.071) | (0.067) | (0.068) | (0.039) | (0.077) |
| 6. Decile | 0.110 | 0.145* | 0.164** | 0.107 | 0.105 | −0.000 | 0.107** | 0.144* |
| Mean: 0.026 | (0.084) | (0.083) | (0.076) | (0.077) | (0.077) | (0.072) | (0.045) | (0.084) |
| 7. Decile | 0.095 | 0.110 | 0.103 | −0.020 | 0.004 | −0.098 | 0.097* | 0.204** |
| Mean: 0.078 | (0.093) | (0.094) | (0.085) | (0.082) | (0.082) | (0.075) | (0.051) | (0.087) |
| 8. Decile | 0.076 | 0.095 | 0.097 | 0.008 | 0.016 | −0.106 | 0.084* | 0.203** |
| Mean: 0.149 | (0.095) | (0.096) | (0.082) | (0.081) | (0.080) | (0.073) | (0.051) | (0.090) |
| 9. Decile | 0.169 | 0.164 | 0.194** | 0.076 | 0.080 | −0.000 | 0.123** | 0.258** |
| Mean: 0.266 | (0.113) | (0.115) | (0.091) | (0.087) | (0.087) | (0.085) | (0.057) | (0.103) |
| 10. Decile: | 0.307* | 0.273* | 0.387*** | 0.208* | 0.204* | 0.093 | 0.208** | 0.385*** |
| Mean: 0.640 | (0.166) | (0.166) | (0.115) | (0.107) | (0.107) | (0.105) | (0.081) | (0.138) |
| Decile 10— | 0.673*** | 0.704*** | 0.783*** | 0.535*** | 0.514*** | 0.288* | 0.235** | 0.314*** |
| Decile 1 | (0.183) | (0.185) | (0.167) | (0.156) | (0.155) | (0.150) | (0.095) | (0.098) |
| Second half— | 0.359*** | 0.385*** | 0.416*** | 0.234* | 0.213* | 0.006 | 0.112* | 0.180** |
| First half | (0.138) | (0.139) | (0.132) | (0.125) | (0.124) | (0.115) | (0.060) | (0.070) |
| Spearman | 0.939*** | 0.939*** | 0.939*** | 0.855*** | 0.855*** | 0.304 | 0.939*** | 0.782*** |
| Correlation | (0.000) | (0.000) | (0.000) | (0.002) | (0.002) | (0.393) | (0.000) | (0.008) |

This table reports the mean monthly returns and the standard errors (in parentheses) for deciles of mutual funds over the period 1984 to 2003 sorted according to the lagged 1-year return gap with back-testing as suggested by Mamaysky, Spiegel, and Zhang (2005). Mutual funds are sorted into deciles according to the average return gaps between 15 and 4 months prior to the portfolio formation. In addition, funds are considered only if the sign of the average return gap equals the sign of the excess reported fund return during the 3 months prior to the portfolio formation. In the first column we report the mean lagged return gap for the decile portfolios after back-testing. We use the performance measures described in Table 4. The returns are expressed in percent per month. The significance levels are denoted by *, **, and *** and indicate whether the results are statistically different from zero at the 10-, 5-, and 1-percent significance levels.

**Table 6**
**Trading strategies using different portfolio formation methods**

| | Return gap (ascending) | | Expenses (descending) | | Holdings return (ascending) | |
|---|---|---|---|---|---|---|
| | Without back-testing | With back-testing | Without back-testing | With back-testing | Without back-testing | With back-testing |
| 1. Decile | −0.199*** | −0.328*** | −0.098* | −0.262*** | −0.049 | −0.216* |
| | (0.062) | (0.099) | (0.052) | (0.097) | (0.123) | (0.125) |
| 2. Decile | −0.123** | −0.209** | −0.125** | −0.270*** | −0.021 | −0.096 |
| | (0.054) | (0.089) | (0.055) | (0.098) | (0.097) | (0.097) |
| 3. Decile | −0.061 | −0.098 | −0.092 | −0.274** | −0.058 | −0.105 |
| | (0.049) | (0.086) | (0.062) | (0.111) | (0.077) | (0.097) |
| 4. Decile | −0.066 | −0.127* | −0.093 | −0.220** | −0.047 | −0.085 |
| | (0.048) | (0.076) | (0.058) | (0.097) | (0.057) | (0.072) |
| 5. Decile | −0.059 | 0.011 | −0.079 | −0.189* | −0.047 | −0.061 |
| | (0.050) | (0.071) | (0.058) | (0.100) | (0.049) | (0.069) |
| 6. Decile | −0.012 | 0.107 | −0.051 | 0.049 | −0.046 | 0.003 |
| | (0.049) | (0.077) | (0.051) | (0.083) | (0.045) | (0.063) |
| 7. Decile | −0.069 | −0.020 | −0.014 | 0.083 | −0.089* | −0.083 |
| | (0.057) | (0.082) | (0.053) | (0.085) | (0.046) | (0.062) |
| 8. Decile | −0.083 | 0.008 | −0.060 | −0.016 | −0.092* | −0.029 |
| | (0.052) | (0.081) | (0.046) | (0.075) | (0.052) | (0.065) |
| 9. Decile | −0.019 | 0.076 | −0.029 | −0.005 | −0.127* | −0.056 |
| | (0.056) | (0.087) | (0.036) | (0.072) | (0.069) | (0.082) |
| 10. Decile | 0.025 | 0.208* | 0.038 | 0.074 | −0.067 | 0.036 |
| | (0.071) | (0.107) | (0.036) | (0.069) | (0.103) | (0.116) |
| Decile 10— | 0.225*** | 0.535*** | 0.135*** | 0.337** | −0.018 | 0.248 |
| Decile 1 | (0.054) | (0.156) | (0.047) | (0.147) | (0.188) | (0.186) |
| Second half— | 0.070*** | 0.234* | 0.074*** | 0.280* | −0.040 | 0.088 |
| First half | (0.024) | (0.125) | (0.022) | (0.153) | (0.094) | (0.108) |
| Spearman | 0.649** | 0.855*** | 0.915*** | 0.818*** | −0.656** | 0.879*** |
| correlation | (0.042) | (0.002) | (0.318) | (0.004) | (0.039) | (0.001) |

This table reports the abnormal monthly returns using the four-factor model of Carhart (1997), along with their standard errors (in parentheses), for deciles of mutual funds formed according to different sorting criteria over the period 1984 to 2003. Funds are sorted on the basis of the lagged return gap, the lagged expense ratio, and the lagged holdings return. For each measure we report, in addition, the performance using the back-testing technique suggested by Mamaysky, Spiegel, and Zhang (2005), by considering only funds where the performance measures of the various criteria are consistent with the excess reported fund return during the 3 months prior to the portfolio formation. The returns are expressed in percent per month. The significance levels are denoted by *, **, and *** and indicate whether the results are statistically different from zero at the 10-, 5-, and 1-percent significance levels.

**Table 7**
**Determinants of the return gap**

| | Dependent variables (in % per month) | | | |
|---|---|---|---|---|
| | Raw return gap | | Abnormal four-factor return gap | |
| Trading costs per month | −0.754*** | −0.826*** | −0.792** | −0.865** |
| | (0.243) | (0.269) | (0.311) | (0.359) |
| Weight of recent IPOs | 0.232*** | 0.243*** | 0.203*** | 0.227*** |
| | (0.027) | (0.030) | (0.032) | (0.037) |
| Correlation between holdings and investor returns | 0.706** | 0.692** | 1.085*** | 1.109*** |
| | (0.314) | (0.333) | (0.361) | (0.391) |
| Expenses per month | −0.339* | −0.276 | −0.237 | −0.216 |
| | (0.180) | (0.187) | (0.238) | (0.226) |
| Turnover | 0.009 | 0.007 | −0.018 | −0.024 |
| | (0.011) | (0.012) | (0.018) | (0.020) |
| Log of TNA | −0.012*** | −0.024*** | −0.014*** | −0.025*** |
| | (0.003) | (0.004) | (0.004) | (0.005) |
| Log of family TNA | | 0.013 *** | | 0.010** |
| | | (0.003) | | (0.004) |
| Log of age | −0.015*** | −0.010 | 0.009 | 0.015 |
| | (0.005) | (0.006) | (0.008) | (0.009) |
| New money growth | 0.433*** | 0.422 ** | 0.674*** | 0.650** |
| | (0.148) | (0.165) | (0.251) | (0.289) |
| New money growth squared | −0.353* | −0.350 | −0.451 | −0.423 |
| | (0.197) | (0.217) | (0.345) | (0.380) |
| Standard deviation of investor returns | 0.011 | 0.014 | 0.015 | 0.004 |
| | (0.010) | (0.011) | (0.010) | (0.010) |
| Load fund indicator variable | −0.005 | −0.018** | 0.006 | −0.000 |
| | (0.007) | (0.008) | (0.010) | (0.012) |
| Index fund indicator variable | −0.041** | −0.047*** | −0.058*** | −0.064*** |
| | (0.016) | (0.016) | (0.020) | (0.022) |
| Size score | −0.032*** | −0.038*** | −0.040*** | −0.042*** |
| | (0.012) | (0.013) | (0.014) | (0.016) |
| Value score | −0.014 | −0.013 | −0.001 | 0.014 |
| | (0.020) | (0.023) | (0.018) | (0.021) |
| Momentum score | −0.067 ** | −0.076** | −0.114*** | −0.125*** |
| | (0.030) | (0.036) | (0.035) | (0.043) |
| Time-fixed effects | Yes | Yes | Yes | Yes |
| Number of observations | 167,983 | 145,328 | 117,130 | 97,788 |
| *R*-squared (in %) | 1.64 | 1.56 | 1.69 | 1.62 |

This table reports the coefficients of the Prais-Winsten panel regressions of the monthly return gaps on various fund and fund family characteristics. The sample includes equity mutual funds and spans the period 1984–2003. The return gap is defined as the difference between the investor fund return and the return based on the previous holdings. All regressions include time-fixed effects and are performed at a monthly frequency. The standard errors (in parentheses) take into account clustering by time. The returns are expressed in percent per month. The significance levels are denoted by *, **, and *** and indicate whether the results are statistically different from zero at the 10-, 5-, and 1-percent significance levels.

## 5.3 Table V and Table VI: it is manager skill, and it earns alpha

**Read:**

- Table V repeats the cyclical-skill exercise at the manager level and then with manager fixed effects. The recession effect survives almost unchanged.
- This is evidence that the phenomenon is not just about the organization or the fund family.
- Table VI then asks whether the identified strategy-switchers actually outperform.
- They do: the `Top` funds earn about 6.8 basis points per month of CAPM alpha, 4.0 basis points of three-factor alpha, and 5.8 basis points of four-factor alpha.
- Annualized, that is roughly 50 to 80 basis points per year of outperformance before expenses and on a risk-adjusted basis.

## 5.4 Table VII and the discussion that follows: who the skilled funds are and how they time

**Read:**

- Table VII shows that the skilled funds are smaller, younger, more expensive, and much more active.
- Relative to other funds, the `Top` funds have about $400 million less TNA, expense ratios higher by 0.26 percentage points, turnover higher by about 50 percentage points, and stronger inflows.

- Their managers are slightly younger and less experienced, and they are much more likely to leave later for hedge funds.
- The text after the table explains *how* they time: they raise cash in recessions, lower portfolio beta, and rotate toward defensive industries.
- This is the section that makes the market-timing result tangible rather than purely statistical.

## 5.5 Table VIII and Table IX: the real-time skill index is investable and predictive

**Table 8**
**Portfolio returns after adjusting for expenses and trading costs**

| | Carhart alpha after expenses | Expenses | Carhart alpha before expenses | Trading costs | Carhart alpha before expenses and trading costs |
|---|---|---|---|---|---|
| 1. Decile | $-0.199^{***}$ | 0.109 | −0.091 | 0.066 | −0.010 |
| | (0.062) | | (0.062) | | (0.062) |
| 2. Decile | $-0.123^{**}$ | 0.099 | −0.024 | 0.047 | 0.027 |
| | (0.054) | | (0.054) | | (0.054) |
| 3. Decile | −0.061 | 0.093 | 0.033 | 0.039 | 0.078 |
| | (0.049) | | (0.049) | | (0.049) |
| 4. Decile | −0.066 | 0.088 | 0.022 | 0.033 | 0.061 |
| | (0.048) | | (0.048) | | (0.049) |
| 5. Decile | −0.059 | 0.086 | 0.027 | 0.031 | 0.064 |
| | (0.050) | | (0.050) | | (0.050) |
| 6. Decile | −0.012 | 0.086 | 0.074 | 0.031 | $0.109^{**}$ |
| | (0.049) | | (0.049) | | (0.049) |
| 7. Decile | −0.069 | 0.089 | 0.020 | 0.036 | 0.063 |
| | (0.057) | | (0.057) | | (0.057) |
| 8. Decile | −0.083 | 0.094 | 0.011 | 0.042 | 0.056 |
| | (0.052) | | (0.052) | | (0.053) |
| 9. Decile | −0.019 | 0.100 | 0.081 | 0.053 | $0.135^{**}$ |
| | (0.056) | | (0.056) | | (0.057) |
| 10. Decile | 0.025 | 0.112 | $0.137^{*}$ | 0.076 | $0.216^{***}$ |
| | (0.071) | | (0.071) | | (0.072) |
| Decile 10— | $0.225^{***}$ | $0.003^{***}$ | $0.227^{***}$ | $0.010^{***}$ | $0.226^{***}$ |
| Decile 1 | (0.054) | (0.001) | (0.054) | (0.001) | (0.053) |
| Second half— | $0.070^{***}$ | $0.001^{***}$ | $0.071^{***}$ | $0.005^{***}$ | $0.072^{***}$ |
| First half | (0.024) | (0.000) | (0.024) | (0.000) | (0.024) |
| Spearman | $0.649^{**}$ | 0.170 | $0.661^{**}$ | 0.171 | $0.697^{**}$ |
| correlation | (0.042) | (0.638) | (0.038) | (0.637) | (0.025) |

This table reports the means and the standard errors (in parentheses) of monthly abnormal returns, expenses, and estimated trading costs for deciles of mutual funds sorted according to the lagged 1-year return gap over the period 1984 to 2003. The return gap is lagged for one additional quarter to account for the possible delay in reporting the holdings. The return gap is defined as the difference between the investor fund return and the return based on the previous holdings. We use the four-factor alpha of Carhart (1997) to measure fund performance. The returns are expressed in percent per month. The table also reports the differences in the return gaps between the top and the bottom deciles and the top and the bottom halves, along with the Spearman rank correlations and the corresponding *p*-values in parentheses. The significance levels are denoted by *, **, and *** and indicate whether the results are statistically different from zero at the 10-, 5-, and 1-percent significance levels.

**Table 9**
**Portfolio returns based on the return gap: various sorting criteria**

| | Return gap 1 Year | Return gap 3 Years | Return gap 5 Years | Return gap subtracting expenses | Return gap for no-load funds | Return gap for nonindex funds |
|---|---|---|---|---|---|---|
| 1. Decile | $-0.199^{***}$ | $-0.234^{***}$ | $-0.146^{**}$ | $-0.221^{***}$ | $-0.183^{***}$ | $-0.203^{***}$ |
| (lowest RG) | (0.062) | (0.060) | (0.068) | (0.062) | (0.068) | (0.062) |
| 2. Decile | $-0.123^{**}$ | $-0.100^{*}$ | $-0.115^{*}$ | $-0.135^{**}$ | $-0.131^{**}$ | $-0.128^{**}$ |
| | (0.054) | (0.055) | (0.063) | (0.053) | (0.059) | (0.054) |
| 3. Decile | −0.061 | −0.060 | $-0.097^{*}$ | −0.057 | −0.072 | −0.070 |
| | (0.049) | (0.049) | (0.054) | (0.051) | (0.058) | (0.050) |
| 4. Decile | −0.066 | −0.029 | $-0.112^{**}$ | −0.074 | −0.004 | −0.073 |
| | (0.048) | (0.043) | (0.049) | (0.052) | (0.059) | (0.049) |
| 5. Decile | −0.059 | $-0.090^{**}$ | −0.053 | −0.057 | −0.027 | −0.056 |
| | (0.050) | (0.046) | (0.050) | (0.050) | (0.063) | (0.052) |
| 6. Decile | −0.012 | −0.055 | $-0.087^{*}$ | −0.013 | −0.000 | −0.019 |
| | (0.049) | (0.050) | (0.049) | (0.050) | (0.052) | (0.051) |
| 7. Decile | −0.069 | −0.075 | −0.100 | −0.085 | −0.008 | −0.072 |
| | (0.057) | (0.057) | (0.061) | (0.054) | (0.062) | (0.058) |
| 8. Decile | −0.083 | −0.067 | −0.013 | −0.067 | $-0.109^{*}$ | −0.086 |
| | (0.052) | (0.057) | (0.066) | (0.051) | (0.058) | (0.052) |
| 9. Decile | −0.019 | 0.019 | 0.038 | 0.012 | −0.078 | −0.015 |
| | (0.056) | (0.060) | (0.069) | (0.053) | (0.057) | (0.056) |
| 10. Decile | 0.025 | −0.017 | 0.012 | 0.028 | 0.029 | 0.026 |
| (highest RG) | (0.071) | (0.067) | (0.070) | (0.071) | (0.079) | (0.072) |
| Decile 10— | $0.225^{***}$ | $0.217^{***}$ | $0.158^{***}$ | $0.249^{***}$ | $0.212^{***}$ | $0.229^{***}$ |
| Decile 1 | (0.054) | (0.056) | (0.054) | (0.057) | (0.081) | (0.055) |
| Second half— | $0.070^{***}$ | $0.063^{***}$ | $0.074^{***}$ | $0.084^{***}$ | 0.050 | $0.072^{***}$ |
| First half | (0.024) | (0.023) | (0.023) | (0.024) | (0.031) | (0.024) |
| Spearman | $0.649^{**}$ | $0.697^{**}$ | $0.879^{***}$ | $0.736^{**}$ | 0.515 | $0.721^{**}$ |
| correlation | (0.042) | (0.025) | (0.001) | (0.015) | (0.128) | (0.019) |

This table reports the monthly abnormal returns according to the four-factor model of Carhart (1997), along with their standard errors (in parentheses), for deciles of mutual funds formed according to different sorting criteria over the period 1984 to 2003. The returns are expressed in percent per month. The significance levels are denoted by *, **, and *** and indicate whether the results are statistically different from zero at the 10-, 5-, and 1-percent significance levels.

**Read:**

- Table VIII sorts funds into quintiles by Skill Index. Performance rises monotonically from Q1 to Q5.
- The Q5–Q1 monthly spread is about 0.512 in abnormal returns, 0.519 in CAPM alpha, 0.293 in three-factor alpha, and 0.257 in four-factor alpha one month after formation.
- The spreads decay over time, but they remain positive even at the 12-month horizon.
- Table IX runs the cleaner predictive regressions with controls. A one-standard-deviation increase in Skill Index predicts about 2.4% higher annual CAPM alpha one month ahead, and about 1.1% to 1.2% higher annual three-factor and four-factor alpha.
- The one-year-ahead predictive results are similarly strong. This is the paper's strongest investor-facing result.

## 5.6 Figure 1: Timing and Picking do not persist, but general skill does

**Read:**

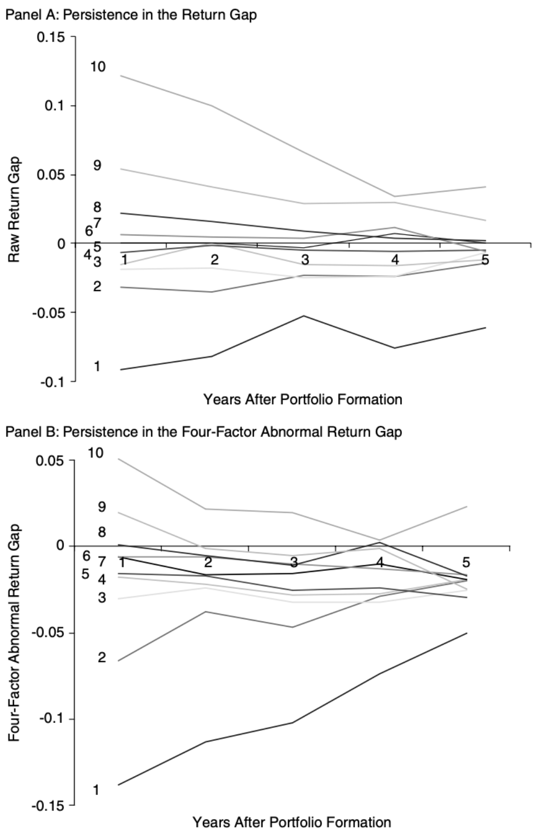


**Figure 1**
**Persistence of the return gap**
This figure depicts the average monthly return gap of portfolios tracked over a 5-year period between 1984 and 2003. The return gap is defined as the difference between the net investor return and the holdings return of the portfolio disclosed in the previous period and is expressed in percent per month. The portfolios are formed by sorting all the funds into deciles according to their initial return gap during the previous year. Subsequently, each portfolio is tracked over the next 5-year period. In Panel A, we report the raw return gap, and in Panel B we report the return gap adjusted for the four-factor Carhart (1997) model.

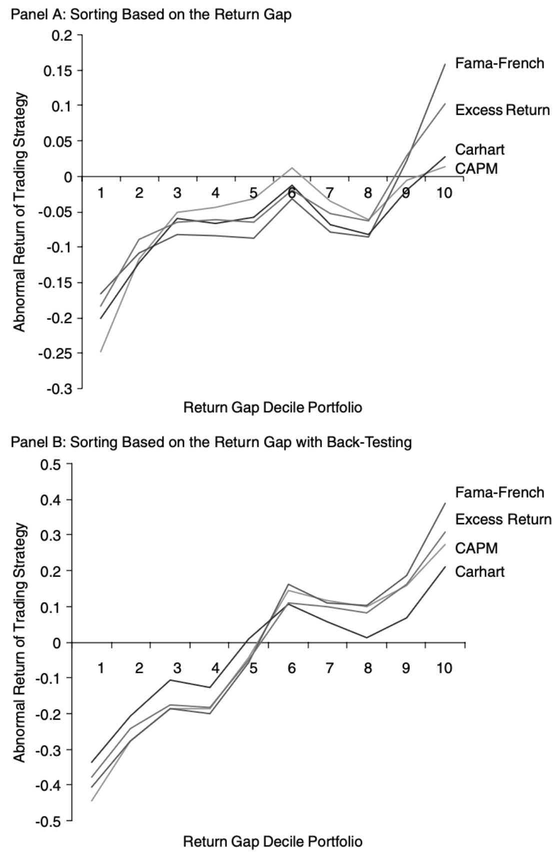


**Figure 2**
**Returns of trading strategies**
This figure shows the average monthly abnormal returns following the formation period over the period between 1984 and 2003, expressed in percent per month. The decile portfolios are formed on the basis of the previous 1-year return gap (Panel A) and on the previous 1-year return gap using the back-testing technique of Mamaysky, Spiegel, and Zhang (2005) (Panel B), in which decile 1 has the lowest return gap and decile 10 has the highest return gap. We use four measures of abnormal returns—the return in excess of the market return; the market-adjusted abnormal return (CAPM); the three-factor adjusted return as in Fama and French (1993); and the four-factor-adjusted return as in Carhart (1997).

- The top panel shows that past Timing has almost no persistence.
- The middle panel shows that past Picking also has almost no persistence.
- The bottom panel shows that the Skill Index *does* persist. High-index managers continue to score better than low-index managers for up to 12 months, and the difference is statistically significant for about six months.
- This is exactly what the paper's theory predicts: task-specific manifestations of skill need not persist if the task changes with the macro state, but general ability should persist.

# 6 Short summary

- The paper measures mutual-fund skill using holdings-based Timing and Picking rather than unconditional alpha alone.
- It finds that skill is state dependent: managers look more like stock pickers in booms and more like market timers in recessions.
- The upper tail of managers matters most, and the same funds that pick well in expansions time well in recessions.
- These funds outperform other funds by about 50 to 80 basis points per year on a risk-adjusted basis.
- A real-time Skill Index built from recession-weighted Timing and Picking is persistent and predicts future net performance.
- The main message is that skill is not one fixed action. It is a flexible information-processing ability whose best use changes over the business cycle.

# 7 Conclusion

## 7.1 Core conclusion (what the paper establishes)

- The paper establishes that mutual-fund skill is time varying in its *expression*. Skilled managers do not always look like stock pickers or always look like market timers.
- Instead, they pick stocks in good times and time the market in bad times.
- A real-time measure built from that state dependence predicts future performance better than either stock picking or market timing by itself.

## 7.2 Economic intuition

- Recessions are times when aggregate shocks matter more and when the payoff to learning about the macroeconomy is higher.
- Expansions are times when firm-specific differentiation matters relatively more.
- A manager with limited attention but high general cognitive ability should therefore reallocate effort across tasks.

- That is exactly the pattern the data reveal.

## 7.3 Takeaway

- This is one of the cleanest papers showing that "manager skill" is not well summarized by a single unconditional alpha number.
- If one wants to evaluate active managers, one should ask not only whether they outperform, but *when* and *how* they outperform.
- The practical lesson is that a manager's general ability may be easier to detect with a state-contingent, flexible index than with any one fixed metric of stock picking or market timing.

## 7.4 Other tables

**Table 10**
**Portfolio returns based on the return gap conditional on market impact**

| | Market impact quintiles | | | | | |
|---|---|---|---|---|---|---|
| | 1. Quintile (lowest market impact) | 2. Quintile | 3. Quintile | 4. Quintile | 5. Quintile (highest market impact) | Quintile 5–Quintile 1 |
| Mean relative trade size (in %) | 0.025 | 0.081 | 0.173 | 0.350 | 1.026 | 1.000 |
| Mean return gap | −0.002 | 0.004 | 0.012 | 0.015 | −0.011 | −0.009 |
| (in % per month) | (0.009) | (0.012) | (0.013) | (0.012) | (0.015) | (0.015) |
| Abnormal return | −0.048 | $-0.097^{**}$ | −0.057 | −0.060 | −0.091 | −0.044 |
| All funds | (0.030) | (0.043) | (0.049) | (0.057) | (0.068) | (0.055) |
| 1. RG quintile | $-0.162^{***}$ | $-0.169^{***}$ | $-0.153^{**}$ | $-0.133^{*}$ | −0.135 | 0.027 |
| | (0.040) | (0.052) | (0.071) | (0.072) | (0.090) | (0.089) |
| 2. RG quintile | 0.015 | $-0.118^{**}$ | −0.043 | −0.073 | $-0.133^{*}$ | $-0.148^{*}$ |
| | (0.038) | (0.046) | (0.054) | (0.065) | (0.080) | (0.075) |
| 3. RG quintile | −0.041 | 0.015 | −0.014 | −0.047 | −0.111 | −0.070 |
| | (0.037) | (0.050) | (0.063) | (0.073) | (0.070) | (0.062) |
| 4. RG quintile | −0.042 | $-0.132^{**}$ | −0.003 | −0.041 | −0.038 | 0.005 |
| | (0.046) | (0.053) | (0.062) | (0.065) | (0.079) | (0.074) |
| 5. RG quintile | −0.012 | −0.052 | −0.004 | 0.048 | 0.003 | 0.016 |
| | (0.050) | (0.062) | (0.060) | (0.083) | (0.089) | (0.082) |
| Quintile 5–Quintile 1 | $0.150^{***}$ | $0.118^{**}$ | $0.148^{**}$ | $0.181^{**}$ | $0.138^{*}$ | −0.012 |
| | (0.055) | (0.056) | (0.072) | (0.074) | (0.084) | (0.105) |
| Quintile 5–Quintile 1 | 0.153 | $0.247^{*}$ | 0.234 | $0.365^{*}$ | $0.534^{***}$ | $0.381^{**}$ |
| With back-testing | (0.123) | (0.140) | (0.165) | (0.192) | (0.203) | (0.163) |

This table reports the monthly abnormal returns according to the four-factor model of Carhart (1997), along with their standard errors (in parentheses), for quintiles formed according to the relative trade size and for quintiles formed according to the lagged return gap between 15 and 4 months prior to the portfolio formation over the period 1984 to 2003. The relative trade size is defined for each mutual fund as the weighted average of the ratio between the absolute value of a trade and the market capitalization of the corresponding stock. The average is weighted by the absolute value of all the trade transactions of a fund. The last row reports the abnormal return difference between the top and the bottom return gap quintiles using the back-testing technique suggested by Mamaysky, Spiegel, and Zhang (2005), by considering only funds in which the performance measures of the various criteria are consistent with the excess reported fund return during the 3 months prior to the portfolio formation. The returns are expressed in percent per month. The significance levels are denoted by *, **, and *** and indicate whether the results are statistically different from zero at the 10-, 5-, and 1-percent significance levels.

**Table 11**
**Portfolio returns based on the return gap conditional on unidentified holdings**

| | Quintiles according to the relative difference in TNAs | | | | | |
|---|---|---|---|---|---|---|
| | 1. Quintile (lowest difference) | 2. Quintile | 3. Quintile | 4. Quintile | 5. Quintile (highest difference) | Quintile 5–Quintile 1 |
| Mean difference in TNAs (in %) | 0.70 | 2.35 | 4.56 | 8.28 | 26.68 | 25.98 |
| Mean return gap | 0.013 | 0.015 | 0.012 | 0.012 | −0.027 | $-0.039^{*}$ |
| (in % per month) | (0.009) | (0.010) | (0.010) | (0.012) | (0.021) | (0.020) |
| Abnormal return | −0.062 | $-0.085^{*}$ | $-0.101^{*}$ | −0.069 | −0.050 | 0.013 |
| All funds | (0.054) | (0.051) | (0.053) | (0.049) | (0.050) | (0.059) |
| 1. RG quintile | −0.111 | $-0.174^{**}$ | $-0.289^{***}$ | $-0.135^{**}$ | $-0.138^{*}$ | −0.028 |
| | (0.074) | (0.069) | (0.067) | (0.068) | (0.071) | (0.088) |
| 2. RG quintile | −0.059 | −0.030 | $-0.120^{**}$ | −0.094 | −0.016 | 0.043 |
| | (0.059) | (0.059) | (0.056) | (0.060) | (0.067) | (0.078) |
| 3. RG quintile | 0.006 | −0.041 | −0.018 | −0.012 | −0.081 | −0.087 |
| | (0.058) | (0.060) | (0.061) | (0.056) | (0.060) | (0.064) |
| 4. RG quintile | −0.019 | $-0.108^{*}$ | −0.035 | −0.081 | −0.029 | −0.009 |
| | (0.064) | (0.058) | (0.066) | (0.062) | (0.058) | (0.071) |
| 5. RG quintile | −0.071 | −0.015 | −0.019 | 0.026 | 0.010 | 0.081 |
| | (0.072) | (0.066) | (0.074) | (0.066) | (0.089) | (0.091) |
| Quintile 5–Quintile 1 | 0.039 | $0.159^{**}$ | $0.270^{***}$ | $0.160^{**}$ | $0.148^{**}$ | 0.109 |
| | (0.066) | (0.069) | (0.070) | (0.069) | (0.080) | (0.096) |
| Quintile 5–Quintile 1 | $0.362^{**}$ | $0.342^{**}$ | $0.577^{***}$ | $0.455^{***}$ | $0.383^{**}$ | 0.025 |
| With back-testing | (0.165) | (0.163) | (0.159) | (0.175) | (0.171) | (0.128) |

This table reports the monthly abnormal returns according to the four-factor model of Carhart (1997), along with their standard errors (in parentheses), for quintiles formed according to the percentage absolute deviation between the disclosed equity holdings and the TNA and for quintiles formed according to the lagged return gap between 15 and 4 months prior to the portfolio formation over the period 1984 to 2003. The last row reports the abnormal return difference between the top and the bottom return gap quintiles using the back-testing technique suggested by Mamaysky, Spiegel, and Zhang (2005), by considering only funds where the performance measures of the various criteria are consistent with the excess reported fund return during the 3 months prior to the portfolio formation. The returns are expressed in percent per month. The significance levels are denoted by *, **, and *** and indicate whether the results are statistically different from zero at the 10-, 5-, and 1-percent significance levels.

**Table 12**
**Predictability of future returns: regression evidence**

| | Dependent variable: abnormal Carhart four-factor returns (in % per month) | | | | |
|---|---|---|---|---|---|
| Prior-year return gap | $0.151^{***}$ | $0.216^{***}$ | $0.213^{***}$ | $0.189^{***}$ | $0.178^{***}$ |
| | (0.035) | (0.044) | (0.045) | (0.048) | (0.051) |
| Prior-year expenses | | $-1.499^{***}$ | $-1.672^{***}$ | $-1.242^{***}$ | $-1.034^{**}$ |
| | | (0.489) | (0.464) | (0.352) | (0.399) |
| Prior-year excess | | $0.141^{**}$ | $0.144^{**}$ | $0.153^{**}$ | $0.157^{**}$ |
| holdings return | | (0.067) | (0.068) | (0.074) | (0.077) |
| Log of lagged TNA | | | $-0.021^{**}$ | $-0.024^{***}$ | $-0.034^{***}$ |
| | | | (0.010) | (0.007) | (0.012) |
| Log of lagged family | | | | | 0.011 |
| TNA | | | | | (0.007) |
| Log of age | | | $-0.022^{*}$ | $-0.024^{*}$ | −0.013 |
| | | | (0.012) | (0.013) | (0.015) |
| Prior-year turnover | | | −0.047 | 0.023 | 0.021 |
| | | | (0.034) | (0.026) | (0.029) |
| Index fund indicator | | | −0.014 | −0.019 | −0.012 |
| variable | | | (0.032) | (0.034) | (0.035) |
| Load fund indicator | | | −0.016 | −0.020 | $-0.034^{*}$ |
| variable | | | (0.015) | (0.017) | (0.019) |
| Trading costs per month | | | | $-2.183^{***}$ | $-2.172^{***}$ |
| | | | | (0.619) | (0.701) |
| Weight of recent IPOs | | | | $0.238^{***}$ | $0.264^{***}$ |
| | | | | (0.067) | (0.077) |
| Correlation between | | | | 0.210 | 0.180 |
| returns | | | | (0.300) | (0.316) |
| New money growth | | | | −0.037 | 0.045 |
| | | | | (0.673) | (0.762) |
| New money growth squared | | | | −0.431 | −0.580 |
| | | | | (0.817) | (0.919) |
| Standard deviation of | | | | −0.012 | −0.008 |
| investor returns | | | | (0.047) | (0.050) |
| Size score | | | | $-0.081^{**}$ | $-0.080^{**}$ |
| | | | | (0.036) | (0.039) |
| Value score | | | | 0.002 | 0.024 |
| | | | | (0.075) | (0.086) |
| Momentum score | | | | $-0.184^{*}$ | $-0.221^{*}$ |
| | | | | (0.097) | (0.113) |
| Time-fixed effects | Yes | Yes | Yes | Yes | Yes |
| Number of observations | 150,946 | 150,946 | 150,210 | 142,083 | 120,969 |
| $R$-Squared (in %) | 8.33 | 8.88 | 9.01 | 9.42 | 9.72 |

This table reports the coefficients of Prais-Winsten regressions of monthly abnormal returns on various fund attributes. The sample includes all equity mutual funds in our sample and spans the period 1984–2003. The dependent variable is the four-factor abnormal return of Carhart (1997). All regressions include time fixed effects and are performed at a monthly frequency. Cluster-corrected standard errors have been provided in parentheses. The returns are expressed in percent per month. The significance levels are denoted by *, **, and *** and indicate whether the results are statistically different from zero at the 10-, 5-, and 1-percent significance levels.